Jibin Johny

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Education

BTech **College of Engineering Chengannur**, Electronics and Communication Engineering *2023 – 2026*

Summary

Embedded Hardware Design Engineer with experience in circuit design, schematic development, PCB design and layout, hardware testing, validation, and debugging. Skilled in analog and digital circuit design, component selection, circuit simulation, prototype testing, and hardware troubleshooting. Proficient in KiCad and LTspice, with a focus on developing reliable and efficient electronic systems. Currently working as a PCB/Hardware Design Engineer in Bengaluru, Karnataka, and seeking opportunities to contribute to innovative hardware development projects while further enhancing technical expertise.

Experience

Hardware Engineer Intern, Aronlabz Tech Pvt Ltd *Sept 2026 – Present*

- Designed, brought up, and validated rugged embedded hardware systems across full product lifecycles.
- Designed schematics and PCB layouts for testing and peripheral boards, including component selection and routing optimization.
- Designed and executed thermal and signal integrity testing using lab instrumentation .

Embeded system Intern, TECHMAGHI *June 2025*

- Programmed wireless robotic car control systems during an intensive hands-on workshop-style internship. Worked with motor control logic and wireless communication protocols under mentorship.

Embeded system Intern, ThinkfoTech *Jan 2023 – Feb 2023*

- Built and tested a Smart Water Tank Monitoring System using IoT sensors and microcontrollers, enabling automated water-level tracking and reducing manual monitoring effort.
- Collaborated on end-to-end system design from sensor wiring to data logging.

Projects

Solar-Powered Public Address Board | KiCad, ESP32 *Oct 2026 – Nov 2026*

- Designed an ESP32-based public address PCB in KiCad with LT3652 solar charging for a single-cell Li-ion battery, DW01A/FS8205A battery protection, and a regulated 3.3 V supply.
- Integrated an I2S MEMS microphone (INMP441), an I2S class-D amplifier (MAX98357A), a GPS UART interface, and USB-C programming through a CH340C with auto-reset.
- Laid out the board for EMI/EMC with a compact charger loop, an antenna keepout, the microphone separated from the amplifier and inductor, and ground pours, then resolved ERC/DRC violations against PCBPower design rules.

Custom ESP32-S3 Development Board | KiCad, ESP32-S3 *Oct 2026 – Nov 2026*

- Designed a custom ESP32-S3 PCB integrating an ESP32-S3-WROOM-1 module, USB-C interface, AMS1117-3.3V regulator, BOOT circuitry, and status LED.
- Developed the schematic and PCB layout, including component selection, footprint assignment, component placement, trace routing, and design rule checking (DRC).
- Implemented USB D+/D- routing, external GPIO and power headers, and an antenna keep-out zone to support reliable RF operation.

Skills

Electronics: Function Generators, Oscilloscopes, Circuit Design & Analysis, Soldering, MCU, SPI/I2C ,UART

Software: KiCad, LTSpice | **Programming:** MATLAB, C/C#/C++, Python